

Certificate of Analysis

PRODUCT DETAILS

Name

Date

Cell Source and Passage

Autologous Activated Natural Killer cells

Storage Temperature

4 – 8°C

Viable Cell Count

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Morphology of Cells	Round to elongated polymorphic shape	Mostly round cells	QUALITY CONTROL APPROVED
Cell Viability	≥90% ± 0.1 of requested cells	Within range	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Immunophenotyping	Positive Marker – CD16/CD56 Negative Marker – CD3	Detected	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Authorised Signature:

Quality Control Department

Certificate of Analysis

Supporting Information

MORPHOLOGY OF CELLS ¹

Figure 1: Natural Killer cells

STERILITY TEST ^{6, 8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.

Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{2,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{3,6}

Negative findings for mycoplasma species *M. hyorhins*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

IMMUNOPHENOTYPING ^{4,7}

Marker	Positive (CD16/56+)	Negative (CD3+)
Result	+	+

‘+’ indicates the presence of the marker in the cell.



REFERENCES

1. Paul W. E. (2013). Fundamental Immunology (7th ed.). Wolters Kluwer Health/Lippincott Williams & Wilkins.
2. Bacterial Endotoxins. European Pharmacopoeia 5.0, <https://gmpua.com/Validation/Method/LAL/EUPHARMACOPEDIA.pdf>
3. US Pharmacopeia <63>. Mycoplasma Tests.
4. William G. Morice. 2007. The Immunophenotypic Attributes of NK Cells and NK-Cell Lineage Lymphoproliferative Disorders.
5. USP Method: Sterilization of Medical Devices, Microbiological Methods.
6. ICH Harmonised Tripartite Guideline, Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q8B.
7. Brynsson YT, Fauriat C, Nunes JM, Wood SN, Björkstén NK, Long EO, Ljunggren HG. Functional analysis of human NK cells by flow cytometry. Methods Mol Biol. 2010;612:335-52. doi: 10.1007/978-1-60761-362-6_23. PMID: 20033652. PMID: PMC4869952.
8. US Pharmacopeia <71>. Sterility Test

